Heterogeneity in Human AI Safety Labeling

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Abstract

Human annotation is widely used to construct benchmarks and evaluate the safety of conversational AI systems, yet the resulting labels may depend systematically on who performs the evaluation. Using the DICES-990 dataset, we examine whether raters' educational attainment predicts safety judgments across four dimensions: overall safety, harmful content, unfair bias, and misinformation. Our results reveal significant educational heterogeneity whose direction varies across safety dimensions. Raters with at most a high-school education are more likely to flag content as broadly unsafe, harmful, or misinformative, whereas college-educated raters are more likely to identify unfair bias. The reversal across categories suggests that education is associated not simply with greater or lower safety sensitivity, but with differences in which types of harm raters identify. Therefore, the demographic composition of an evaluator pool can influence measured AI safety and should be treated as an explicit design choice in benchmark construction and safety auditing.

CCS Concepts

• **Computing methodologies → Machine learning & Artificial Intelligence**.

Keywords

AI safety evaluation; human raters; educational heterogeneity; DICES dataset; autoraters; LLM-as-judge

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1 Introduction

The alignment of large language models (LLMs) with human values requires reliable ground truth about what constitutes "safe" or "harmful" content. That ground truth has historically been produced by human annotators: workers who read AI-generated conversations and label them across a range of safety dimensions. The implicit assumption is that human judgment, aggregated over enough raters, converges on a stable and accurate signal. This assumption is increasingly contested.

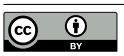
A growing body of evidence shows that safety perceptions are not universal but are instead shaped by the social position, cultural background, and demographic characteristics of the person doing the evaluation [1, 3, 11]. What one rater identifies as dangerous misinformation, another may regard as a legitimate opinion. What one person experiences as unfair bias, another may not even notice. These disagreements are not random noise—they are systematically related to who the rater is [4, 10].

Education is a particularly consequential axis of heterogeneity. Educational attainment shapes exposure to formal frameworks for reasoning about harm, fairness, and epistemic accuracy [2]. Higher-educated individuals may be more attuned to particular conceptions of bias—especially those prominent in academic discourse—while lower-educated individuals may be more sensitive to certain direct or practical harms. Either way, the direction of disagreement need not be uniform across safety categories, which is precisely what makes education such a revealing variable for studying rater heterogeneity.

This heterogeneity has practical consequences for AI development. Human safety labels are used not only to describe model behavior but also to construct evaluation benchmarks, compare candidate models, and provide feedback for alignment. If identical model outputs receive systematically different safety labels depending on the composition of the evaluator pool, then measured safety can partly reflect evaluator composition rather than model behavior alone. Consequently, rater recruitment and aggregation are not merely implementation details: they are components of the evaluation design. Understanding which dimensions of rater heterogeneity affect safety judgments can therefore inform the construction of more representative annotation panels, demographic sensitivity analyses, and hybrid evaluation systems that combine heterogeneous human judgment with standardized automated assessment.

This paper contributes to this debate in three ways. First, we use the DICES-990 dataset [1] to provide systematic empirical evidence that educational attainment predicts safety labeling in a large, diverse, and purpose-built dataset of adversarial human–AI conversations. Second, we show that the direction of the education effect is not uniform across safety categories: for overall safety, harmful content, and misinformation, less-educated raters flag more content; for unfair bias, the opposite is true. Third, we derive practical implications for AI safety evaluation: evaluator-pool composition should be reported and stress-tested as part of benchmark design, and standardized automated raters can be used as complements to human evaluation when consistency across evaluation rounds is important.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 describes the data and empirical



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strategy. Section 4 presents our findings. Section 5 concludes with implications for AI safety evaluation methodology.

2 Literature Review

Human feedback is the cornerstone of reinforcement learning from human feedback (RLHF), the dominant paradigm for aligning LLMs with human preferences [9]. RLHF presupposes that human judgments, once aggregated, represent a coherent normative standard. But as Aroyo et al. [1] argue, this presupposition is empirically questionable: "safety is both socially and culturally situated," and treating it as a binary property obscures meaningful variation across human populations.

Recent empirical work has confirmed this concern across multiple domains. Rastogi et al. [11] provide evidence that raters from different racial and ethnic groups diverge systematically in their safety assessments of multimodal AI outputs. Gordon et al. [4] show that inter-rater disagreement is not merely measurement error but reflects coherent and legitimate differences in how humans reason about harm and fairness. Pavlick and Kwiatkowski [10] make a related point in the context of natural language inference, arguing that some apparent annotation "errors" are actually principled reflective judgments. More recently, a study of expert clinicians evaluating AI-generated mental health responses found inter-rater reliability consistently below acceptable thresholds, with disagreement reflecting "coherent but incompatible individual clinical frameworks" rather than noise [5].

Education shapes both factual knowledge and normative frameworks, and there is growing evidence that it mediates perceptions of AI risk and safety. Vesely and Kim [13] find that socioeconomic variables, including educational attainment, are correlated with public support for AI safety oversight, though the direction of effects varies across specific regulatory questions. Capraro et al. [2] review a broad literature on the socioeconomic drivers of AI-related beliefs, noting that education tends to amplify engagement with AI-related risks that are framed in terms of systemic bias and distributional harm—categories with particular salience in academic discourse.

The scalability problem in human evaluation has motivated the development of automated raters—LLMs that evaluate other LLMs' outputs [15]. The "LLM-as-a-judge" paradigm has demonstrated promising results: Lambert et al. [6] show that reward models can serve as reliable proxies for human preferences in many evaluation tasks, and Mazeika et al. [8] demonstrate that automated safety classifiers achieve strong agreement with human raters on safety-relevant content.

More recent work has pushed this further. Xie et al. [14] show that fine-tuned small LLMs (7B parameters) can achieve over 80% agreement with human annotations for safety refusal evaluation, performing comparably to GPT-4o at a fraction of the computational cost. Luo et al. [7] propose memory-augmented reasoning frameworks that allow LLMs to emulate expert human evaluators in agent safety assessment tasks. These advances suggest that autoraters are becoming technically mature enough to serve as the primary evaluation mechanism in many settings.

The normative case for autoraters is also growing. Zheng et al. [15] argue that autoraters offer consistency advantages over human raters precisely because they do not vary by demographic background. While autoraters have their own biases—including tendencies toward verbosity, self-enhancement, and position effects [12]—these biases are, in principle, systematically characterizable and correctable in ways that human demographic heterogeneity is not. The key insight is not that autoraters are perfect, but that human raters are systematically imperfect in ways that undermine the foundational premise of human-as-gold-standard.

3 Data and Methodology

3.1 Data

Our analysis uses the DICES-990 dataset [1], released by Google Research as an open benchmark for conversational AI safety evaluation. The dataset contains 990 multi-turn adversarial conversations between human participants and a conversational AI model. Each conversation was rated by a diverse pool of 173 raters recruited from the United States and India, yielding approximately 60–70 unique safety annotations per conversation across 24 safety criteria—far exceeding the replication rates of typical annotation projects. The rater pool was explicitly constructed to achieve demographic balance across gender, age cohort, and racial/ethnic group, making it well-suited for the kind of heterogeneity analysis we conduct.

Our primary outcomes of interest are binary safety labels derived from four questions: (1) *Q_overall*: whether the AI's response is overall unsafe [1]; (2) *Q2_harmful_content_overall*: whether the response contains harmful content; (3) *Q3_unfair_bias_overall*: whether the response exhibits unfair bias; and (4) *Q4_misinformation_overall*: whether the response spreads misinformation. Each outcome is coded 1 if the rater answered "Yes" and 0 if they answered "No."

Our primary predictor is rater educational attainment, a binary variable distinguishing raters with a college degree or higher ("College+") from those with at most a high-school diploma ("High-School–").

Figure 2 presents raw mean "Yes" rates by education group for each of the four safety outcomes. Each bar shows the mean proportion of ratings in which the rater labeled the content as unsafe, with 95% confidence intervals computed via the standard error of the mean.

Two patterns stand out. First, for overall safety, harmful content, and unfair bias, College+ raters flag more content than High-School– raters. Second, this direction reverses for misinformation: College+ raters flag more content as exhibiting misinformation. These raw differences motivate the regression analysis below.

3.2 Regression Specification

To isolate the education effect from potential confounders, we estimate a series of logistic regression models of the form:

$$\Pr(Y_{ic} = 1) = \Lambda\big(\beta_0 + \beta_1 \,\text{HighSchool}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \delta_p\big), \qquad (1)$$

[1] Overall unsafety is the aggregated DICES rating Q_overall, equal to 1 if the rater answered "Yes" to any granular safety sub-question and 0 if the rater answered "No" to all the questions.

Figure 1: Mean Safety Label Rates by Education Group

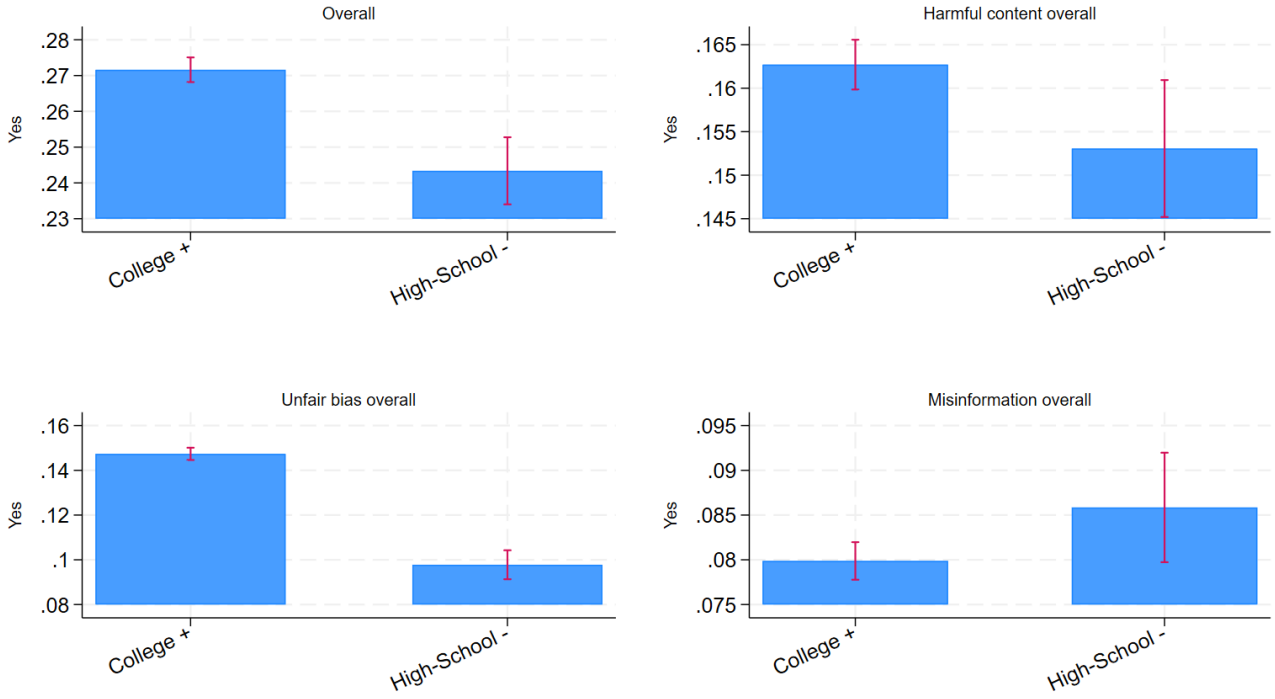


Figure 2: Bars show mean "Yes" rates with 95% confidence intervals; panels are overall safety, harmful content, unfair bias, and misinformation.

where Y_{ic} is the binary safety label assigned by rater i to conversation c; $\Lambda(\cdot)$ is the logistic CDF; HighSchool_i is an indicator equal to 1 for raters with at most a high-school education; $\mathbf{X}_i$ is a vector of rater-level controls including gender, age (linear), and locale (US vs. India); and δ_p is a phase fixed effect to account for the fact that conversations were rated in multiple annotation rounds. Standard errors are heteroskedasticity-robust throughout. All four safety outcomes are estimated using this common specification, allowing direct comparisons.

4 Empirical Results

4.1 Empirical Results

Table 1 presents logistic regression estimates of the effect of educational attainment on four safety labeling outcomes, controlling for gender, location, and phase fixed effects across samples of roughly 68,000 to 70,000 rater-conversation observations. The High-School coefficient is positive and statistically significant for three of the four outcomes — overall safety (0.071, $p < 0.05$), harmful content (0.067, $p < 0.10$), and misinformation (0.180, $p < 0.01$) — indicating that raters with at most a high-school education are more likely than college-educated raters to flag content as unsafe, harmful, or misinformative. The reversal is most pronounced for unfair bias, where the High-School coefficient is negative and highly significant (-0.278, $p < 0.01$), confirming that college-educated raters are substantially more likely to identify unfair bias in AI-generated content. Age is positive and significant across all four models, suggesting that older raters are consistently more likely to apply safety labels regardless of dimension, with the largest age effect appearing in the unfair bias model (0.293, $p < 0.01$). The models fit reasonably well given the complexity of the labeling task, with chi-squared statistics significant at $p < 0.001$ in all four cases, though the pseudo-R^2 values (0.010–0.031) indicate that demographic characteristics explain a modest but meaningful share of labeling variance. Collectively, these results establish that educational attainment is a robust, statistically significant predictor of safety labeling behavior, and that its direction depends critically on which safety dimension is under consideration — a pattern consistent with education shaping not simply overall risk sensitivity, but the specific normative frameworks through which raters recognize and categorize harm.

Figure 4 displays the predicted probabilities of a "Yes" safety label at each education level, derived from our logistic regression models and computed at mean values of all control variables. The vertical axis in each panel is the predicted probability that a rater labels the respective content dimension as unsafe.

Table 1: Heterogeneity in AI Safety Labeling

	Overall		Harmful content		Unfair Bias		Misinformation	
	Coef.	se	Coef.	se	Coef.	se	Coef.	se
High-School	0.071**	(0.029)	0.067*	(0.035)	-0.278***	(0.044)	0.180***	(0.046)
Age	0.159***	(0.013)	0.154***	(0.016)	0.293***	(0.017)	0.132***	(0.022)
Fixed Effects:								
Gender	Yes		Yes		Yes		Yes	
Location	Yes		Yes		Yes		Yes	
Phase	Yes		Yes		Yes		Yes	
N	68352		69619		69567		70245	
Chi2	1291.885		844.794		1588.096		337.222	
p	0		0		0		0	
r2_p	0.016		0.014		0.031		0.01	

*, **, *** denote statistical significance at 10%, 5%, and 1%, respectively. Standard errors are reported in parentheses.

Figure 3: Predicted Probability of Safety Labeling by Rater Education

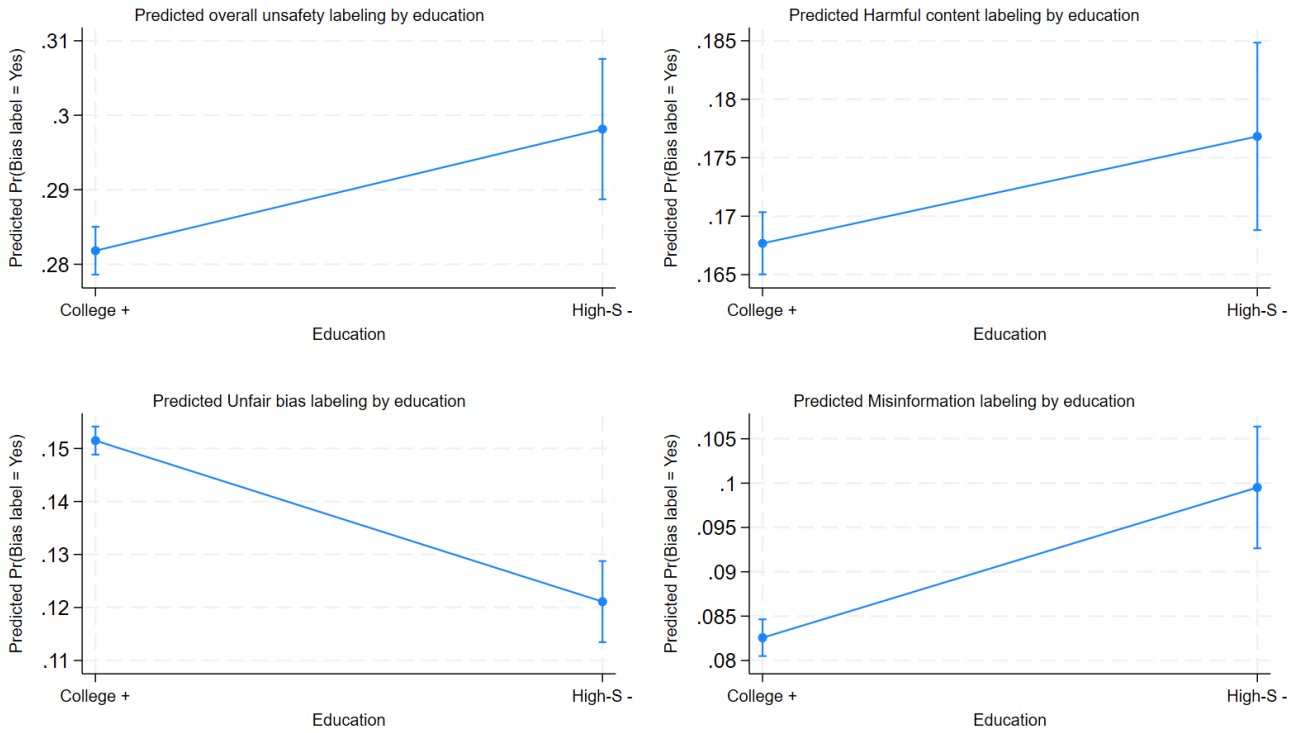


Figure 4: Predicted probabilities of a "Yes" safety label by rater education, from logistic regression models with controls for gender, age, locale, and phase fixed effects. Dots show point estimates; capped lines show 95% confidence intervals. Panels correspond to overall safety (top-left), harmful content (top-right), unfair bias (bottom-left), and misinformation (bottom-right).

The results reveal a consistent and statistically discernible pattern of educational heterogeneity, with the direction of effects varying across safety dimensions. We discuss each in turn.

College+ raters assign a predicted probability of approximately 0.282 for flagging content as unsafe overall, compared to approximately 0.295 for High-School– raters. This difference of roughly

1.3 percentage points, while modest in absolute terms, is consistent with the direction seen in the descriptive statistics. High-school-educated raters are somewhat more likely to flag content as broadly unsafe, a pattern that may reflect lower familiarity with contextualizing AI-generated outputs within broader discursive or rhetorical conventions.

The education gradient for harmful content follows a similar direction. College+ raters have a predicted probability of approximately 0.166 of flagging content as harmful, versus approximately 0.175 for High-School− raters. The difference is modest but in the same direction as overall safety, suggesting that lower-educated raters are marginally more sensitive to perceived direct harms in AI outputs.

The most striking result concerns the unfair bias dimension, where the direction of the education effect reverses sharply. College+ raters have a predicted probability of approximately 0.150 of flagging content as exhibiting unfair bias, compared to only approximately 0.119 for High-School− raters—a gap of roughly 3 percentage points. This reversal is substantively meaningful and statistically robust. It suggests that formal education equips raters with a particular sensitivity to bias-coded content, likely reflecting the prominence of concepts such as structural discrimination and representational harm in academic and professional discourse. Less-educated raters, by contrast, appear less likely to invoke an "unfair bias" label even when the same content is rated by college-educated peers as biased.

For misinformation, the education gradient is smaller and the confidence intervals are wider, indicating greater uncertainty. The predicted probability of a misinformation label is approximately 0.081 for College+ and 0.095 for High-School− raters, a difference that is in the same direction as overall safety and harmful content but is less precisely estimated. This may reflect the relatively lower base rate of misinformation labels in the data.

These findings paint a picture of systematic educational heterogeneity in safety labeling. The reversal of the education effect between harmful-content and unfair-bias dimensions is especially informative. It suggests that education does not simply make raters more or less "safety-conscious" in a general sense; rather, it shapes which *kinds* of harms raters are sensitized to. College-educated raters bring frameworks from academic and professional training that emphasize bias and representational harm, while less-educated raters may respond more readily to content they perceive as directly harmful or deceptive in everyday contexts.

This finding has direct implications for AI safety evaluation methodology. If the composition of a rater pool—particularly its educational distribution—systematically shapes the safety labels it produces, then the "ground truth" generated by human annotation is not a universal standard but a reflection of the particular pool employed. Evaluation datasets built from predominantly college-educated annotators, as is common in practice, will systematically underweight certain harm perceptions while overweighting others. Conversely, datasets built from less-educated pools will reverse these biases.

This is not a counsel of despair about human evaluation. It is, rather, a principled argument for complementing human evaluation with automated raters that are not demographically situated in the same way. An autorater does not have an educational background that sensitizes it to some categories of harm and blinds it to others. Its biases, while real, are of a different character—and, crucially, they are stable, systematically characterizable, and in principle correctable [6, 14, 15]. The case for autoraters is therefore not that they are superior to humans, but that the heterogeneity documented here undermines the premise that any particular human rater pool produces an authoritative gold standard.

4.2 Robustness

We tested the robustness of our findings by augmenting the baseline logit specifications with an interaction between age and education. This allows for the possibility that the education effect on safety labeling differs across age cohorts (Gen Z, Millennials, Gen X+). The main patterns described above are qualitatively unchanged: the direction of the education coefficient remains positive for harmful content and negative (i.e., College+ flag more) for unfair bias across all age groups. The interaction terms are not consistently significant, suggesting that while age cohort may introduce additional heterogeneity, it does not fundamentally alter the education gradient documented above.

Table 2 augments the baseline specification with an interaction between High-School education and age, allowing the education gradient in safety labeling to vary across rater age cohorts. The results reveal a richer and more nuanced picture than the additive model. The main effect of High-School becomes negative and highly significant for overall safety (-0.862, $p < 0.01$) and harmful content (-0.513, $p < 0.01$), indicating that among younger raters, lower-educated individuals are actually *less* likely to flag content in these categories than their college-educated peers. The sign reverses for unfair bias, where the High-School main effect is now positive and significant (0.742, $p < 0.01$), while the misinformation coefficient (-0.212) is not statistically distinguishable from zero. The interaction term High-School × Age is positive and significant for overall safety (0.424, $p < 0.01$), harmful content (0.258, $p < 0.01$), and misinformation (0.174, $p < 0.05$), meaning that the propensity of lower-educated raters to flag these safety dimensions increases with age — older raters without a college degree become progressively more likely to apply safety labels relative to younger raters at the same education level. The interaction is negative and highly significant for unfair bias (-0.459, $p < 0.01$), indicating that the elevated bias-flagging tendency of lower-educated raters attenuates sharply as age increases. Age retains its positive and significant main effect across all four models, and model fit improves modestly relative to Table 1, with pseudo-R^2 values rising slightly (0.011–0.032). Together, these interaction results underscore that the education effect on safety labeling is not uniform across the life course: for most safety dimensions, the gap between education groups widens with age, while for unfair bias it narrows — a finding that further complicates any simple characterization of which raters flag more content and why.

5 Conclusion

This paper has used the DICES-990 dataset to demonstrate that educational attainment is a systematic predictor of how human raters evaluate the safety of AI-generated conversations. The direction

Table 2: Heterogeneity in AI Safety Labeling (with Education x Age Interaction)

	Overall		Harmful content		Unfair Bias		Misinformation	
	Coef.	se	Coef.	se	Coef.	se	Coef.	se
High-School	-0.862***	(0.103)	-0.513***	(0.129)	0.742***	(0.129)	-0.212	(0.168)
Age	0.095***	(0.015)	0.117***	(0.017)	0.348***	(0.018)	0.105***	(0.025)
High-School x Age	0.424***	(0.043)	0.258***	(0.053)	-0.459***	(0.057)	0.174**	(0.07)
Fixed Effects:								
Gender	Yes		Yes		Yes		Yes	
Location	Yes		Yes		Yes		Yes	
Phase	Yes		Yes		Yes		Yes	
N	68352		69619		69567		70245	
Chi2	1358.413		853.129		1719.663		361.673	
p	0		0		0		0	
r2_p	0.018		0.014		0.032		0.011	

*, **, *** denote statistical significance at 10%, 5%, and 1%, respectively. Standard errors are reported in parentheses.

of the education effect reverses across safety categories: lower-educated raters are more likely to flag content as broadly unsafe or harmful, while college-educated raters are more likely to identify unfair bias. This heterogeneity is not random noise—it reflects the different normative frameworks that individuals with different educational backgrounds bring to the task of safety evaluation.

These findings reinforce a growing body of evidence showing that AI safety is not a stable property waiting to be read off by neutral observers, but a socially situated judgment that varies systematically with who is doing the judging [1, 4, 10, 11]. The implications for AI development are significant. Safety datasets constructed from homogeneous rater pools will embed the normative sensitivities of those pools into the models trained on them, potentially causing those models to be more responsive to some forms of harm while remaining blind to others.

We have argued that this state of affairs provides a principled motivation for investing in automated raters as scalable, consistent, and demographically neutral complements to human evaluation [6, 7, 14, 15]. Autoraters are not a panacea: they have their own biases and limitations, and they cannot replace human judgment in high-stakes or highly contextual situations. But their biases are of a different character— systematic, stable, and in principle amenable to empirical characterization and correction in ways that demographic heterogeneity in human rater pools is not.

Future work should extend this analysis in several directions. First, intersectional analyses combining education with race, gender, and locale could reveal more complex patterns of heterogeneity than any single demographic variable can capture. Second, the mechanisms underlying educational differences in safety labeling deserve more careful investigation: is the effect driven by critical-thinking skills, content familiarity, normative framework acquisition, or some combination? Third, the practical implications for dataset construction need to be explored: given the heterogeneity documented here, how should rater pool composition be designed to produce safety labels that are broadly representative rather than reflecting the sensitivities of any particular educational stratum?

For practitioners, the implication is concrete: safety evaluations should document the demographic composition of human rater pools, report subgroup-specific results where feasible, and test whether conclusions are robust to plausible changes in evaluator composition. Automated raters may provide a useful standardized reference point, but they should be validated against heterogeneous human judgments rather than treated as inherently unbiased substitutes for them. What this paper establishes is a baseline empirical fact: who rates the AI matters for what gets labeled as unsafe. Designing evaluation systems that are robust to that dependence should therefore be treated as a core component of AI safety methodology.

References

[1] Lora Aroyo, Alex S. Taylor, Mark Díaz, Christopher M. Homan, Alicia Parrish, Greg Serapio-García, Vinodkumar Prabhakaran, and Ding Wang. 2023. DICES Dataset: Diversity in Conversational AI Evaluation for Safety. In *Advances in Neural Information Processing Systems*, Vol. 36. https://arxiv.org/abs/2306.11247 NeurIPS 2023 Datasets and Benchmarks Track.

[2] Valerio Capraro, Austin Lentsch, Daron Acemoglu, Selin Akgun, Aisel Akhmedova, Ennio Bilancini, Jean-François Bonnefon, et al. 2024. The Impact of Generative Artificial Intelligence on Socioeconomic Inequalities and Policy Making. *PNAS Nexus* 3, 6 (2024), pgae191. doi:10.1093/pnasnexus/pgae191

[3] Mark Díaz, Ian D. Kivlichan, Rachel Rosen, Dylan K. Baker, Razvan Amironesei, Vinodkumar Prabhakaran, and Emily Denton. 2022. CrowdWorkSheets: Accounting for Individual and Collective Identities Underlying Crowdsourced Dataset Annotation. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*. ACM, 2342–2351. doi:10.1145/3531146.3534647

[4] Mitchell L. Gordon, Kaitlyn Zhou, Kayur Patel, Tatsunori Hashimoto, and Michael S. Bernstein. 2021. The Disagreement Deconvolution: Bringing Machine Learning Performance Metrics In Line With Reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM. doi:10.1145/3411764.3445423

[5] Kiana Jafari, Paul Ulrich Nikolaus Rust, Duncan Eddy, Robbie Fraser, Nina Vasan, Darja Djordjevic, Akanksha Dadlani, Max Lamparth, Eugenia Kim, and Mykel Kochenderfer. 2026. Expert Evaluation and the Limits of Human Feedback in Mental Health AI Safety Testing. *arXiv preprint arXiv:2601.18061* (2026).

[6] Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2025. RewardBench: Evaluating Reward Models for Language Modeling. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 1755–1797. https://arxiv.org/abs/2403.13787

[7] Hanjun Luo, Shenyu Dai, Chiming Ni, Xinfeng Li, Guibin Zhang, Kun Wang, Tongliang Liu, and Hanan Salam. 2025. AgentAuditor: Human-Level Safety and Security Evaluation for LLM Agents. In *Advances in Neural Information Processing*

Systems. https://arxiv.org/abs/2506.00641 NeurIPS 2025.

[8] Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal. *arXiv preprint* (2024). arXiv:2402.04249.

[9] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training Language Models to Follow Instructions with Human Feedback. In *Advances in Neural Information Processing Systems*, Vol. 35. 27730–27744.

[10] Ellie Pavlick and Tom Kwiatkowski. 2019. Inherent Disagreements in Human Textual Inferences. *Transactions of the Association for Computational Linguistics* 7 (2019), 677–694. doi:10.1162/tacl_a_00293

[11] Charvi Rastogi, Tian Huey Teh, Pushkar Mishra, Roma Patel, Zoe Ashwood, Aida Mostafazadeh Davani, Mark Díaz, Michela Paganini, Alicia Parrish, Ding Wang, Vinodkumar Prabhakaran, Lora Aroyo, and Verena Rieser. 2024. Insights on Disagreement Patterns in Multimodal Safety Perception across Diverse Rater Groups. arXiv:2410.17032 [cs.AI] https://arxiv.org/abs/2410.17032

[12] Rickard Stureborg, Dimitris Alikaniotis, and Yoshi Suhara. 2024. Large Language Models Are Inconsistent and Biased Evaluators. *arXiv preprint* (2024). arXiv:2405.01724.

[13] Stepan Vesely and Byungdoo Kim. 2024. Survey Evidence on Public Support for AI Safety Oversight. *Scientific Reports* 14 (2024), 31491. doi:10.1038/s41598-024-82977-5

[14] Tinghao Xie, Xiangyu Qi, Yi Zeng, Yangsibo Huang, Udari Madhushani Sehwag, Kaixuan Huang, Luxi He, Boyi Wei, Dacheng Li, Ying Sheng, Ruoxi Jia, Bo Li, Kai Li, Danqi Chen, Peter Henderson, and Prateek Mittal. 2025. SORRY-Bench: Systematically Evaluating Large Language Model Safety Refusal. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=YfKNaRktan

[15] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, Vol. 36. https://arxiv.org/abs/2306.05685 NeurIPS 2023.